

NYC Taxi Trip Duration Prediction using Geospatial Feature Engineering and Gradient Boosting

Abstract

Accurate prediction of taxi trip duration is an important problem in urban transportation systems, ride-hailing services, and fleet management operations. Reliable travel time estimates can improve customer experience, optimize driver allocation, and enhance dispatch efficiency.

This project develops a machine learning-based trip duration prediction system using the NYC Taxi Trip Duration dataset. The workflow includes data cleaning, exploratory data analysis, feature engineering, model comparison, feature selection, cross-validation, and hyperparameter optimization. Multiple regression algorithms were evaluated, including Linear Regression, Ridge Regression, Lasso Regression, Decision Tree Regressor, Random Forest Regressor, and Gradient Boosting Regressor.

The final optimized Gradient Boosting model achieved a test R^2 score of 0.763, Mean Absolute Error (MAE) of 207.89 seconds, and Root Mean Squared Error (RMSE) of 316.11 seconds. The analysis revealed that travel distance, pickup hour, travel direction, and weekend status are the most influential factors affecting trip duration.

1. Introduction

Ride-hailing platforms such as Uber and Lyft depend heavily on accurate trip duration estimation for route planning, driver assignment, customer ETA prediction, and fleet optimization.

Predicting trip duration is a challenging problem because travel time is influenced by multiple factors including travel distance, traffic congestion, time of day, day of week, and travel direction.

The objective of this project is to develop a machine learning model capable of accurately predicting NYC taxi trip duration using trip information available at the start of a journey.

2. Problem Statement

The goal of this project is to predict the duration of taxi trips in New York City using trip characteristics such as pickup location, dropoff location, passenger count, and temporal information.

Business Objectives

- Improve ETA estimation
- Optimize taxi dispatch operations
- Improve fleet utilization
- Reduce passenger waiting times
- Support operational planning

Machine Learning Objective

This is a supervised regression problem where:

Target Variable:

trip_duration

Evaluation Metrics:

- R^2 Score
 - Mean Absolute Error (MAE)
 - Root Mean Squared Error (RMSE)
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3. Dataset Description

The NYC Taxi Trip Duration dataset contains information about taxi trips including:

- Vendor information
- Pickup and dropoff timestamps
- Pickup coordinates
- Dropoff coordinates
- Passenger count
- Trip duration

Dataset Overview

Attribute	Description
vendor_id	Taxi vendor identifier

pickup_datetime	Trip start timestamp
dropoff_datetime	Trip end timestamp
passenger_count	Number of passengers
pickup_latitude	Pickup latitude
pickup_longitude	Pickup longitude
dropoff_latitude	Dropoff latitude
dropoff_longitude	Dropoff longitude
store_and_fwd_flag	Store-and-forward indicator
trip_duration	Trip duration in seconds

4. Data Cleaning and Preprocessing

Data quality validation was performed prior to analysis.

Cleaning Steps

- Converted datetime columns into datetime format.
- Validated geographic coordinates.
- Removed invalid pickup and dropoff locations.
- Removed trips shorter than one minute.
- Removed trips longer than two hours.
- Verified absence of missing values and duplicate records.

Dataset Size

Original Dataset Size:

729,322 records

Cleaned Dataset Size:

723,758 records

Percentage Removed:

0.76%

This indicates that the dataset was generally clean and required minimal correction.

5. Exploratory Data Analysis

Target Variable Analysis

Trip duration exhibited a positively skewed distribution.

Statistical characteristics:

- Skewness: 2.20
- Kurtosis: 7.70

A logarithmic transformation produced a more symmetric distribution.

Figure 1: Distribution of Trip Duration

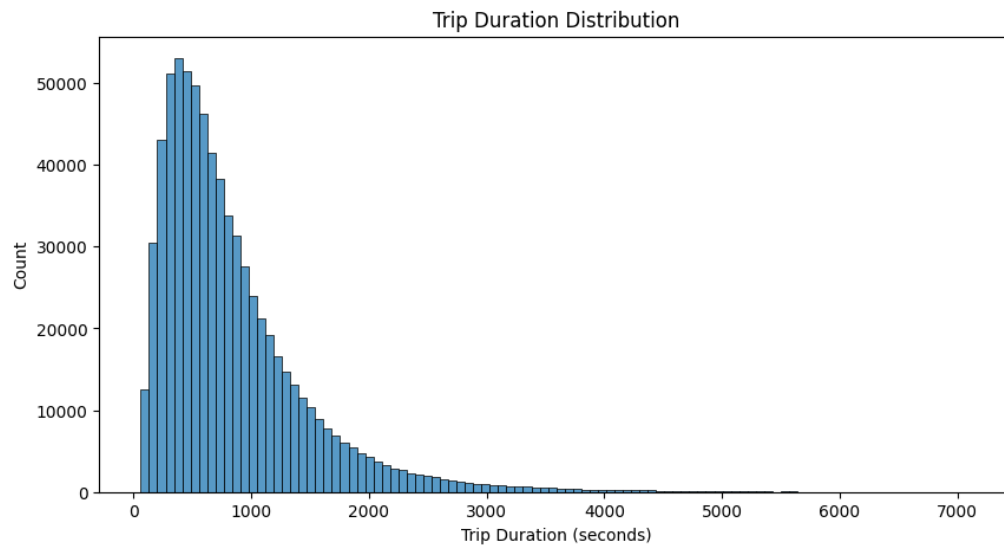
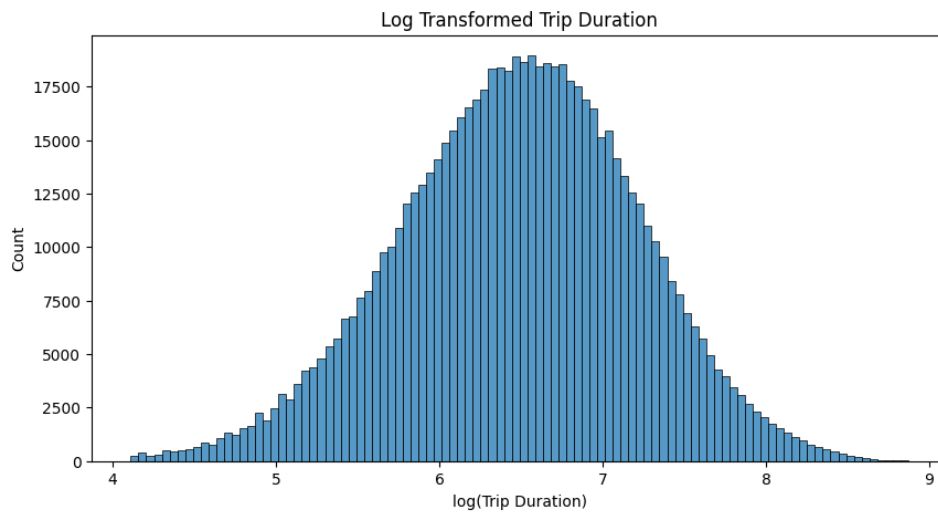


Figure 2: Log-Transformed Trip Duration Distribution



Temporal Analysis

Trip demand varied significantly throughout the day.

Key observations:

- Lowest activity occurred between 3 AM and 5 AM.
- Demand increased during morning commuting hours.
- Peak activity occurred during evening commuting periods.
- Trip durations increased during rush-hour periods due to congestion.

Figure 3: Trips by Pickup Hour

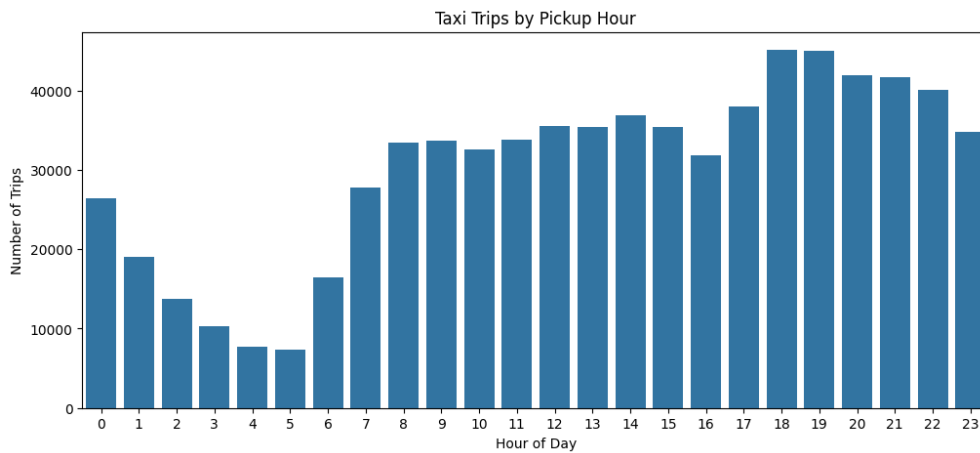
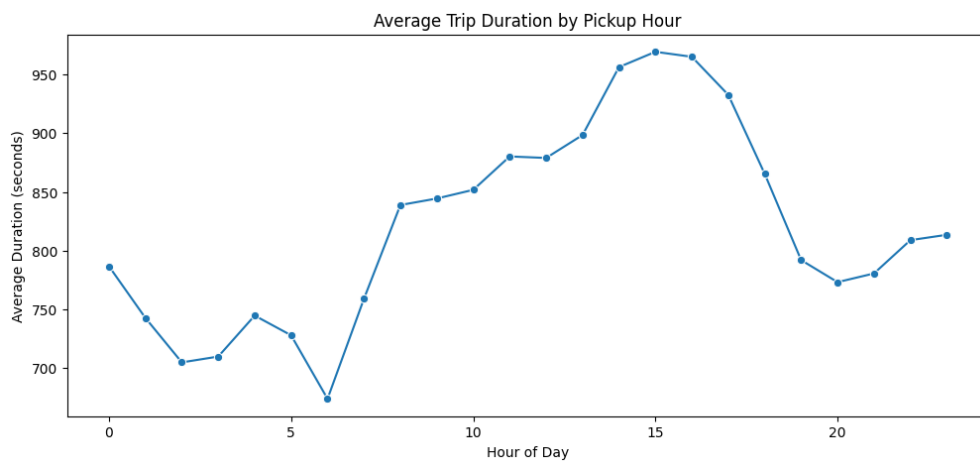


Figure 4: Average Trip Duration by Pickup Hour



Weekday Analysis

Weekday travel patterns differed significantly from weekend travel patterns.

Key observations:

- Higher trip volumes during weekdays.
- Longer average trip durations during weekdays.
- Weekend trips were generally shorter.

Figure 5: Trips by Day of Week

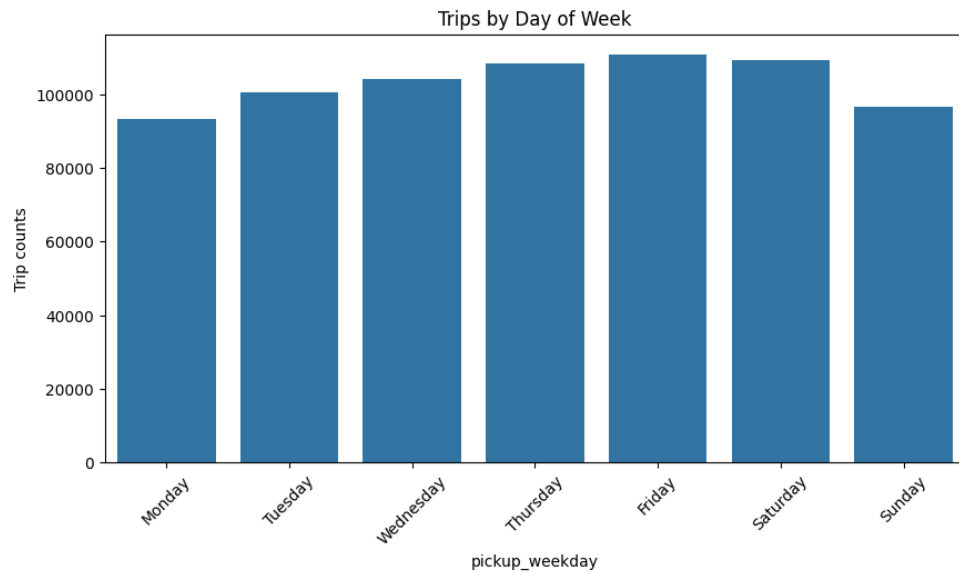
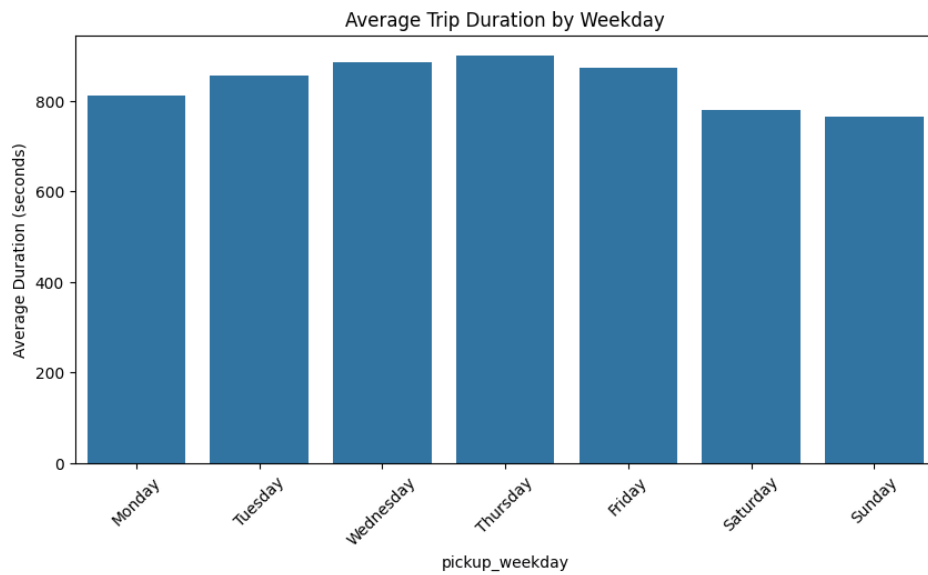


Figure 6: Average Trip Duration by Day of Week



Geographic Analysis

Geospatial analysis was performed to understand the spatial distribution of taxi pickups and drop-offs across New York City.

The visualizations show a strong concentration of taxi activity around Manhattan and surrounding boroughs, highlighting the importance of location-based features in trip duration prediction.

Figure 7: Pickup Location Density Map (with NYC Basemap)

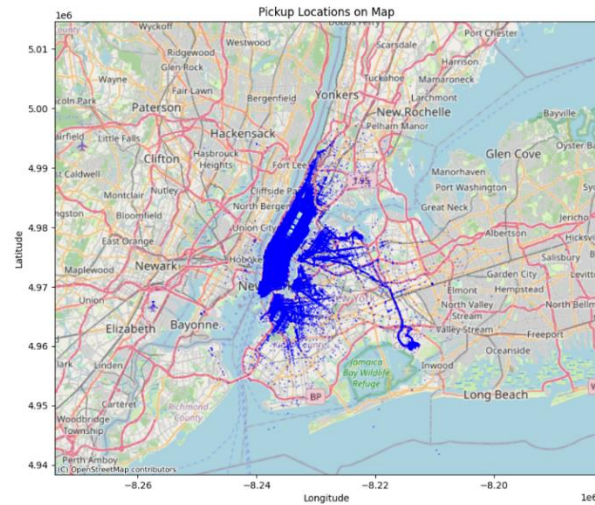
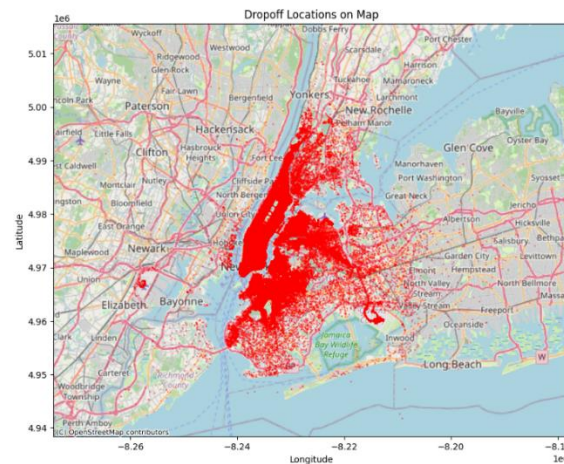


Figure 8: Dropoff Location Density Map (with NYC Basemap)



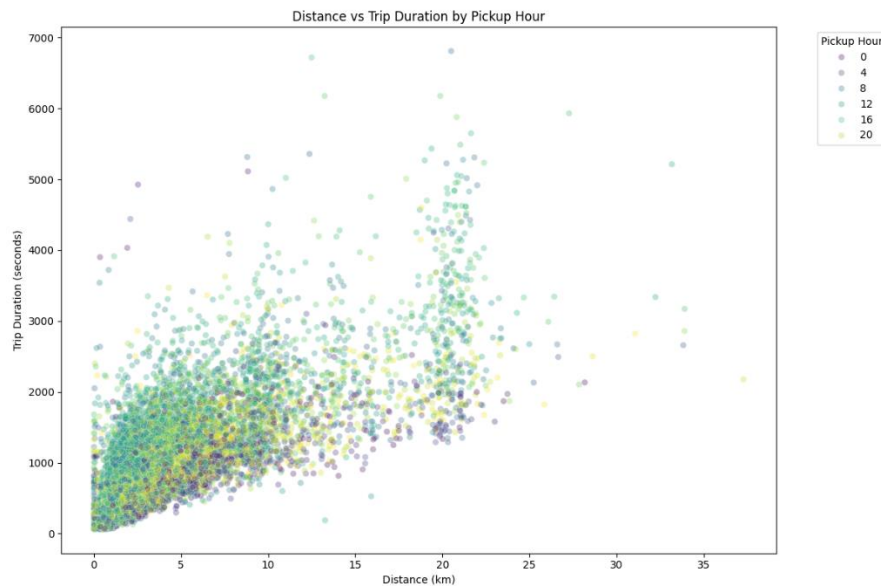
Distance Analysis

The engineered Haversine distance feature showed a strong positive relationship with trip duration.

Correlation: Distance vs Trip Duration = 0.771

The visualization demonstrates that while travel distance is the primary determinant of trip duration, pickup hour introduces additional variability. Trips covering similar distances can experience significantly different travel times depending on traffic conditions throughout the day. This observation supports the inclusion of temporal features in predictive modeling.

Figure 9: Distance vs Trip Duration Scatter Plot



6. Feature Engineering

Several new features were created to improve predictive performance.

Temporal Features

- pickup_hour
- pickup_day
- pickup_month
- pickup_weekday
- is_weekend

Geospatial Features

- Haversine Distance
- Manhattan Distance
- Bearing

Categorical Encoding

- Weekday dummy variables
- Store-and-forward encoding

These engineered variables enabled the model to capture temporal and geographic travel behavior.

7. Model Development

The following models were evaluated:

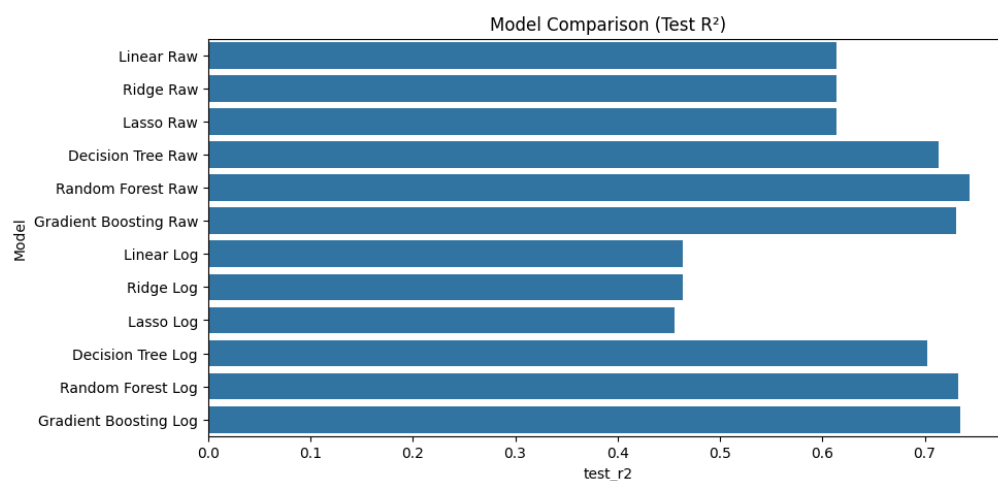
1. Linear Regression
2. Ridge Regression
3. Lasso Regression
4. Decision Tree Regressor
5. Random Forest Regressor
6. Gradient Boosting Regressor

Both raw and log-transformed targets were investigated.

8. Model Evaluation

Model Comparison Results

Figure 10: Model Comparison (Test R^2)



Key findings:

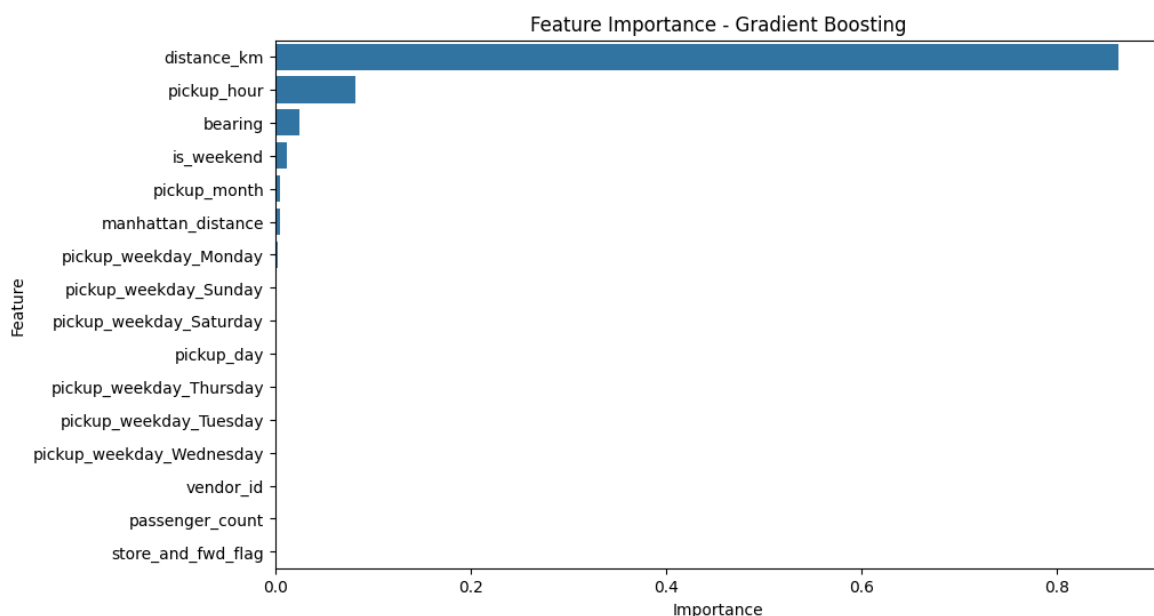
- Tree-based models significantly outperformed linear models.
 - Regularization provided minimal improvement.
 - Random Forest achieved the highest raw accuracy.
 - Gradient Boosting demonstrated superior generalization.
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9. Model Optimization and Validation

Feature Importance Analysis

Feature importance revealed that distance-based and temporal features dominate prediction performance.

Figure 11: Feature Importance Plot



Key findings:

- distance_km accounted for approximately 86% of model importance.
 - pickup_hour was the second most important feature.
 - bearing and is_weekend contributed additional predictive value.
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Feature Selection

A simplified model using only four features:

- distance_km
- pickup_hour
- bearing
- is_weekend

retained approximately 99% of the predictive performance achieved by the full-feature model.

Cross Validation

10-fold cross-validation was performed.

Results:

- Mean R^2 : 0.726
- Standard Deviation: 0.008

These results indicate stable and consistent model performance.

Hyperparameter Tuning

Grid Search was performed on the Gradient Boosting Regressor.

Best Parameters:

- n_estimators = 200
- max_depth = 5
- learning_rate = 0.1

Best Cross-Validation Score: 0.752

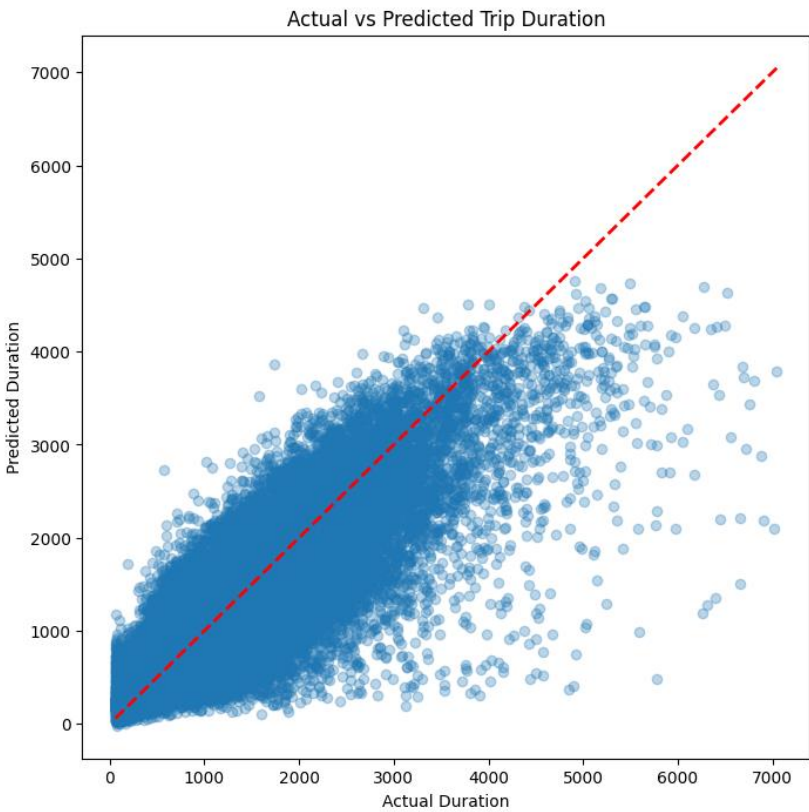
Final Model Performance

The optimized model was retrained using the complete cleaned dataset.

Table 1: Final Model Performance

Metric	Value
Model	Tuned Gradient Boosting Regressor
Training R ²	0.768
Testing R ²	0.763
MAE	207.89 seconds
RMSE	316.11 seconds

Figure 12: Actual vs Predicted Trip Duration



10. Business Insights

Insight 1

Travel distance is the primary driver of trip duration, accounting for approximately 86% of model importance.

Insight 2

Traffic congestion significantly impacts travel time during peak commuting hours.

Insight 3

Weekday trips are generally longer and more frequent than weekend trips.

Insight 4

Travel direction contributes additional predictive information, suggesting route-specific traffic patterns.

Insight 5

A small subset of features captures most predictive information, enabling efficient and interpretable prediction systems.

Insight 6

Machine learning can significantly improve dispatch efficiency and ETA estimation accuracy.

11. Conclusion

This project successfully developed a machine learning system capable of predicting NYC taxi trip duration using temporal and geospatial information.

The study demonstrated the importance of feature engineering, particularly distance-based and time-based features, in improving predictive performance.

Among the evaluated models, the tuned Gradient Boosting Regressor achieved the best balance between predictive accuracy and generalization capability.

Final model performance:

- Test R^2 Score: 0.763
- MAE: 207.89 seconds

- RMSE: 316.11 seconds

The results indicate that machine learning can effectively support transportation analytics and operational decision-making within urban mobility systems.

References

1. NYC Taxi Trip Duration Dataset
2. Scikit-Learn Documentation
3. Python Data Science Libraries Documentation (Pandas, NumPy, Matplotlib, Seaborn)
4. Gradient Boosting Regressor Documentation